

Sahil Shivaji Sawant

Shrewsbury, MA | +1-716-390-2316 | sahilshivajisawant@gmail.com | [LinkedIn](#) | [GitHub](#)

Work Experience

Arta Support

AI Software Engineer

Mar 2026 - May 2026

Remote, USA

- Scaled a prototype **multi-agent** troubleshooting system into an enterprise platform serving four distinct roles, delivering real-time graph-guided responses at **~30s average latency**.
- Engineered a graph-guided **agentic RAG** system combining Pinecone with Neo4j improving retrieval precision by 30% and reducing knowledge base update time through a PDF to Knowledge Graph pipeline.
- Designed **Text-to-Cypher** query prompts for creating graph schemas to enable multi-hop traversal over entities and relationships.
- Deployed a full-stack user authentication and API key management system using **AWS (Cognito, DynamoDB, and KMS)** adding admin approval workflows, role-based dashboards, and self-serve API key rotation.
- Built **token-accounting** infrastructure to meter per-user and per-model LLM token usage across **AWS EC2** hosted services, enabling cross-region budget controls.

University at Buffalo

Research Assistant

Aug 2025 - Jan 2026

Buffalo, USA

- Developed a vision-guided object-following system for the **Unitree Go2** quadruped robot, achieving **48ms** latency and **90%+** target tracking success in indoor environments.
- Created a **voice AI** pipeline for the robot using **Whisper STT**, and a regex-based **NLP parser** dispatching commands over SSH to the on-robot execution loop supporting **20+ distinct behaviors**.
- Trained a perception pipeline using a **quantized YOLOv8n model** deployed on Jetson Nano, trained on a custom-annotated dataset with geometric and photometric augmentations achieving **mAP@0.5 of 0.995**.

Capgemini

Analyst

Dec 2022 - May 2024

Mumbai, India

- Managed job scheduling and deployment pipelines using AppWorx, monitoring 30+ workflow chains per shift across 24-hour rotating operations, promoting chains across Dev, QA, and Production and coordinating cross-team notifications for aborts and crashes to maintain continuous pipeline availability.
- Replaced manual data-handling workflows with optimized PL/SQL packages and stored procedures across Oracle EBS modules, reducing manual intervention by 35% and improving run consistency.
- Performed data reconciliation and validation across EBS batch loads, identifying and correcting data discrepancies before they propagated into dependent job chains.

Projects

DeepSpeech Therapy – Real-Time Voice AI Coach

- Integrated a real-time voice AI coaching system that routes errors from a speech-analysis stack to a **GPT-4o coaching agent**, which delivers context-aware articulation feedback based on detected speech errors.
- Enabled clarity detection in speech sampled using a **1D Convolutional Autoencoder** having **96.89% accuracy** and a **CNN + Transformer** for stutter diagnosis (Block, Prolongation, Repetition)

End-to-End ML Pipeline for Job-Market Classification

- Built an end-to-end ML pipeline over 1.3M+ LinkedIn job postings using **PySpark** on **Databricks**, training a five-class job-category classifier at **87% accuracy**, and deployed it as a real-time Amazon SageMaker endpoint with Model Monitor for data-drift detection.
- Automated retrain-and-redeploy through a **GitHub Actions CI/CD workflow** with model versioning and validation gates with a Streamlit prototype for interactive inference.

DermAI - AI powered dermatology Assistant

- Created a **multimodal** dermatology assistant that orchestrates a **vision ensemble** (ResNet50, DenseNet121, VGG-19, EfficientNet-B0) as a classification tool feeding a **Gemini 3 Flash LLM** generating diagnostic feedback across **7 lesion categories** on HAM10000
- Applied **SMOTE** and **class-weighted** training to address severe **class imbalance**, reaching 94% precision & 0.93 F1 (up from 0.87 single-model) with evaluation comparing single-model vs. ensemble outputs.

Skills

- **Languages & Libraries:** Python, PySpark, SQL/PL-SQL, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Spacy, NLTK
- **GenAI & ML:** LangGraph, LangChain, RAG Systems, Agentic Workflows, MCP, Agent to Agent communication, LLM Orchestration, Prompt Engineering, Hugging Face, CNNs, RNNs, LSTMs, Transformers, Supervised & Unsupervised Learning
- **MLOps & Deployment:** Amazon SageMaker, SageMaker Model Monitor, CI/CD (GitHub Actions), Model Deployment, Model Monitoring / Drift Detection, Model Versioning, Streamlit
- **Backend & Frontend:** FastAPI, Node.js, React, Socket.IO, REST APIs, Git, Linux/Unix
- **Databases & Cloud:** Neo4j, Qdrant, Pinecone, DuckDB, AWS, GCP, Azure, Docker, Apache Spark, Oracle EBS

Education

University at Buffalo

MS, Artificial Intelligence

Aug 2024 - Dec 2025

Buffalo, USA

University of Mumbai

BE, Electronics & Telecommunication

Aug 2018 - May 2022

Mumbai, India